Tommaso Grigoletto

– Postdoctoral fellow –

Birth date: January 2nd 1996
Nationality: Italian
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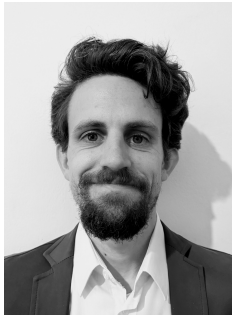
Media Links:



Research statement

My research lies at the intersection of control theory, applied mathematics, and quantum science and technology. In particular, I develop theoretical and computational methods for modelling, analysing, controlling and reducing quantum dynamical systems. More broadly, my research aims to establish systematic control-theoretic frameworks for understanding and engineering the challenges posed by emerging quantum technologies.

Research profile at a glance

- Research on model reduction for quantum dynamics;
- Author of 9 journal papers (6 as first author) including: IEEE Transactions on Automatic Control, IEEE Transactions on Information Theory, Automatica, Quantum;
- Collaborations with Dartmouth College, Université Paul Sabatier, Inria Paris;
- Teaching experience as main instructor in ''Fundamentals of automation engineering'';
- Organizer of the Junior QTech seminar series.

Professional Experience

Research experience

01/07/2025 →30/06/2026 **Postdoctoral Fellow** University of Padua, Padua, Italy
Project: Developing aggregation methods for network model reduction
Advisors: Prof. Mattia Zorzi and Prof. Francesco Ticozzi
Highlight: Developed an approximation technique for deriving time-dependent reduced generators [15].

01/11/2023 →30/06/2025 **Postdoctoral Fellow** University of Padua, Padua, Italy
Project: Approximate reduction methods for classical and quantum networks
Advisor: Prof. Francesco Ticozzi
Highlight: Expanded the algebraic model reduction technique toward quantum trajectories [13] and approximate reductions onto slow manifolds [17].

06/08/2022 →10/02/2023 **Visiting Scholar** Dartmouth College, Hanover, NH (USA)
Project: Exact model reduction for continuous-time quantum dynamics
Advisor: Prof. Lorenza Viola
Highlight: Developed an algebraic model reduction technique for open quantum systems [9].

Teaching experience

Main instructor 2023-2025 — **Fundamentals of Automation Engineering**,
Bachelor's degree in Engineering Physics, Ca'Foscari University
Three consecutive years, 48 hours each year

Teaching assistant 2022-2023 — **Control Engineering Laboratory**,
Master's degree in Control Systems Engineering, University of Padua
Two consecutive years, 26 hours each year

Education

01/10/2020 →30/09/2023 **Ph.D. in Information Engineering**, University of Padua Padua, Italy
Thesis: Exact model reduction for quantum systems
Advisor: Prof. Francesco Ticozzi
Defended March 21st, 2024, **with honors**.

29/09/2018 →21/07/2020 **M.Sc. in Automation Engineering**, University of Padua, Padua, Italy
Thesis: Measurement-based switching control for quantum systems
Advisor: Prof. Francesco Ticozzi
Final grade: 110/110

28/08/2015 →16/07/2018 **B.Sc. in Information Engineering**, University of Padua, Padua, Italy
Thesis: Control system for a rocket soft landing
Advisor: Prof. Mauro Bisiacco
Final grade: 106/110

Sep. 2010 →Jul. 2015 **High school**, I.T.I.S. Rossi, Vicenza, Italy
Thesis: Control system for a quadcopter
Advisor: Prof. Paolo Fumene Ferruglio
Final grade: 94/100

Awards, Fellowships and Grants

2026 **Seal of Excellence – Marie Sklodowska-Curie Action Postdoctoral Fellowship** (92.4%)
Project: ''IMPAQT: Improved Magic state Purity viA Quantum conTrol'' with Prof. Alain Sarlette;
2022 Gini scholarship, Fondazione Aldo Gini, University of Padua;
2020 Ph.D. scholarship, University of Padua;
2015 Tekne scholarship, Regione Veneto;
2014 Leonardo scholarship, Fondazione Centro Produttività Veneto;
2013 Participated to the Robocup World Cup in Eindhoven.

Peer recognition

2026 Paper [14] was selected as Editors' Suggestion for Physical Review A.
2025 I received the IOP Trusted Reviewer status for an 'Outstanding' review report.

Presentations and talks

Conferences

1. December 11th 2025 – **"Krylov spaces and algebras for efficient simulation of quantum dynamics"**, QCSC Workshop 2025, Trento Italy;
2. September 9th 2025 – **"Exact Model Reduction for Continuous-Time Open Quantum Dynamics"**, Poster presentation at IQIS 2025, Bologna Italy;
3. December 2024 – **"Exact Model Reduction for Discrete-Time Conditional Quantum Dynamics"**, Invited session at IEEE Conference on Decision and Control, Milan;
4. December 2022 – **"Minimal resources for exact simulation of quantum walks"**, IEEE Conference on Decision and Control, Cancun ;

Seminars and Lectures

1. March 12th 2026 – **"Model reduction for complex networks (and quantum systems)"**, Prin kick-off meeting, University of Padua;
2. January 26th 2026 – **"Krylov spaces and algebras for efficient simulation of quantum dynamics"**, QSNOW Workshop 2026, Asiago Italy;
3. March 13th 2025 – **"Exact Model Reduction for Quantum Dynamics"**, Invited talk at Quantum Nano Seminar - Dartmouth College;
4. January 20th 2025 – **"Applications of quantum conditional expectations"**, Two-hour lecture included in the Ph.D. course "Quantum Probability Information and Dynamics" held by Francesco Ticozzi, University of Padua;

5. January 10th 2025 – **"Quantum conditional expectation and the Takesaki Theorem"**, Two-hour lecture included in the Ph.D. course "Quantum Probability Information and Dynamics" held by Francesco Ticozzi, University of Padua;
6. September 22nd 2023 –**"Exact model reduction for Quantum Systems: an algebraic approach"**, Invited talk at Inria Paris, Quantic group;
7. June 27th 2023 – **"Exact model reduction for Quantum Systems: an algebraic approach"**, Seminar talk for the quantum lunch at the department of information engineering, University of Padua;
8. April 2022 – **"Algebraic reduction of Hidden Markov Models"**, Seminar talk at the department of information engineering, University of Padua ;

Student supervision

Bachelor Students

2025	Giacomo Girotto Ca'Foscari	"Experiments on a single qubit quantum computer", Supervisor: **T.G.**
2025	Rosanna Mantese Ca'Foscari	"Control of a Boeing v-22 Osprey", Supervisor: **T.G.**, Co-Supervisor: Matthias Pezzutto
2025	Marco Grolla Ca'Foscari	"Control of a rocket soft-landing", Supervisor: **T.G.**, Co-Supervisor: Matthias Pezzutto
2025	Stefano Santello Ca'Foscari	"Analysis and control of a pendulum actuated with reaction wheels", Supervisor: **T.G.** , Co-Supervisor: Matthias Pezzutto
2025	Marco Tesio Ca'Foscari	"Simulation of open quantum systems using quantum computers", Supervisor: **T.G.**
2025	Marco Vanzetto Ca'Foscari	"Parameter estimation for closed quantum systems", Supervisor: **T.G.**
2024	Giuseppe D'Auria UniPD	"Computer based state observers for magnet systems", Supervisor: Damiano Varagnolo, Co-Supervisor: **T.G.**
2024	Jacopo Tomasetig Ca'Foscari	"Modelling and Control of tire-road interaction dynamics", Supervisor: **T.G.**
2024	Alessandra Dal Bello Ca'Foscari	"Tracking visual markers for maglev platforms", Supervisor: **T.G.**, Co-Supervisor: Damiano Varagnolo

Master Students

Ongoing	Mauro Carpanese UniPD	Supervisor: Francesco Ticozzi, Co-supervisor: **T.G.**
Ongoing	Francesco Sartori UniPD	Supervisor: Francesco Ticozzi, Co-supervisor: **T.G.**

International collaborations

- Lorenza Viola, Department of Physics and Astronomy, Dartmouth College;
- Clément Pellegrini, Laboratoire de Statistique et Probabilités, Université Paul Sabatier;
- Alain Sarlette, Quantic group, Inria Paris.

Professional skills

Reviewing, Service and Outreach

- Associate editor for:
  - ECC 2026;
  - IEEE qCCL 2026.
- Reviewed for journal including:
  - IEEE Control System Letters (12 papers),
  - Automatica (3 papers),
  - International Journal of control (1 paper),
  - Journal of Physics A (1 paper).
- Reviewer for conferences:
  IEEE Conference on Decision and Control 2021,2022,2024,2025;
  IEEE American control conference 2025.
- Founder and main organizer of the Junior Qtech seminar cycle: a postdoc led cycle of seminars aimed at merging the skills of physicists, mathematicians, engineers and chemists of the Padua QTech center.

- Co-organizer with Prof. Ticozzi of the Invited session "New tools for estimation, modeling and control of quantum systems" at the Conference on Decision and Control 2024.

Languages

Mother tongue:	Italian
Other languages:	English ($\simeq$ C1 level)

Programming

Advanced:	Python, Matlab, LaTeX
Experience with:	Java, C++

Publications

*Legend: *=Shared first-authorship*

Journal

[2] **T. Grigoletto** and F. Ticozzi. "Stabilization via feedback switching for quantum stochastic dynamics". In: *IEEE Control Systems Letters* 6 (2021), pp. 235–240.

[4] **T. Grigoletto** and F. Ticozzi. "Algebraic Reduction of Hidden Markov Models". In: *IEEE Transactions on Automatic Control* 68.12 (2023).

[5] **T. Grigoletto** and F. Ticozzi. "Model Reduction for Quantum Systems: Discrete-time Quantum Walks and Open Markov Dynamics". In: *IEEE Transactions on Information Theory* (2025).

[6] W. Liang, **T. Grigoletto**, and F. Ticozzi. "Dissipative feedback switching for quantum stabilization". In: *Automatica* 165 (2024), p. 111659.

[7] **T. Grigoletto** and F. Ticozzi. "Exact Model Reduction for Discrete-Time Conditional Quantum Dynamics". In: *IEEE Control Systems Letters* (2024).

[8] M. Cortese, **T. Grigoletto**, F. Ticozzi, and A. Ferrante. "Robust positive model reduction via monotone matrices". In: *IEEE Transactions on Automatic Control* (2025).

[9] **T. Grigoletto**, Y. Tao, F. Ticozzi, and L. Viola. "Exact Model Reduction for Continuous-Time Open Quantum Dynamics". In: *Quantum* 9 (2025), p. 1814.

[13] **T. Grigoletto**, C. Pellegrini, and F. Ticozzi. "Quantum model reduction for continuous-time quantum filters". In: *Annales Henri Poincaré* (2025).

[14] M. Peruzzo, **T. Grigoletto**, and F. Ticozzi. "Reconstructing Quantum States and Expectations via Dynamical Tomography". In: *Physical Review A* (2026). Accepted, yet to appear. Selected as Editors' Suggestion.

Conference

[1] T. Benciolini*, T. Grigoletto*, and M. Zorzi. "Image compression by means of the multidimensional circulant covariance extension problem–Revisited". In: *2020 59th IEEE Conference on Decision and Control (CDC)*. IEEE. 2020, pp. 280–285.

[3] **T. Grigoletto** and F. Ticozzi. "Minimal resources for exact simulation of quantum walks". In: *2022 IEEE 61st Conference on Decision and Control (CDC)*. 2022, pp. 5155–5160.

[11] M. Peruzzo, **T. Grigoletto**, and F. Ticozzi. "Reconstructing Quantum States from Local Observation: A Dynamical Viewpoint". In: *2024 IEEE 63st Conference on Decision and Control (CDC)*. 2024.

[16] **T. Grigoletto**, L. Viola, and F. Ticozzi. "Model reduction for controlled quantum Markov dynamics". In: *European Control Conference (ECC) 2026*. Accepted, yet to appear. 2025.

Under review

[15] **T. Grigoletto**. *Computing time-dependent reduced models for classical and quantum dynamics*. Submitted to Automatica. 2025.

[17] **T. Grigoletto**, A. Sarlette, F. Ticozzi, and L. Viola. *Approximate Reduced Lindblad Dynamics via Algebraic and Adiabatic Methods*. Submitted to Quantum Science ad Technology. 2026.

Other references

[10] **T. Grigoletto**. "Exact model reduction for quantum systems". Ph.D. dissertation. University of Padua, 2024.

[12] **T. Grigoletto**, A. Gallina, E. Bellesso, and G. Pillonetto. *Applied Functional Analysis and Machine Learning*. Lecture notes. 2024.

In preparation

[18] **T. Grigoletto**, M. Casanova, L. Viola, and F. Ticozzi. *Weak symmetries and where to find them*. In preparation.

[19] **T. Grigoletto** and A. Sarlette. *On the well posedness of adiabatic elimination at higher orders*. In preparation.

[20] **T. Grigoletto**, T. Benoist, C. Pellegrini, L. Viola, and F. Ticozzi. *Symmetries for quantum trajectories*. In preparation.